

Clean proof of the basin-expansion proposition with intuition, algorithm, and Lean4 guidance

Notes for Arina

Reading guide

- Section 1 is the cartoon. Read first.
- Section 2 is the algorithm in pseudocode and how each line maps to the symbols used in Section 4.
- Section 3 fixes a single notation. The paper’s appendix has minor drift; this section pins one name per object.
- Section 4 is the elegant three-part proof. Each part uses exactly one classical tool.
- Section 6 is a Lean4 skeleton. Use it as a checklist while writing the human proof: every **sorry** corresponds to a step you must justify in prose.

1 Intuitive picture

Two vector fields live on parameter space $\mathbb{R}^d$:

- the plain policy-gradient field $v(\phi)$, with a known fixed point at the Nash equilibrium ϕ^* ;
- the Meta-MAPG (shaped) field $F_\lambda(\phi) = v(\phi) + \lambda M(\phi)$, parameterised by shaping strength $\lambda \geq 0$.

The proposition compares them. The whole story is three sentences:

- (i) Adding λM *moves the zero*. The new zero ϕ_λ^* is at distance $O(\lambda)$ from ϕ^* . As $\lambda \rightarrow 0$ it slides back.
- (ii) Around the new zero, the shaped field *pulls inward harder* than v pulled toward ϕ^* . Contraction rate goes from μ to $\mu + \lambda\mu_M$.
- (iii) Stronger pull $\Rightarrow$ *wider certified ball* of guaranteed contraction. Radius $\rho_\lambda = (\mu + \lambda\mu_M)/(2L_\lambda)$ instead of $\rho_0 = \mu/(2L_0)$.

Mental model. Think of v as a cone-shaped funnel above ϕ^* . The funnel has a fixed slope μ and a fixed mouth radius ρ_0 . Adding λM bends the funnel: the bottom of the funnel slides sideways by $O(\lambda)$ (the new zero), the slope steepens to $\mu + \lambda\mu_M$ (sharper drift), and the mouth widens to $\rho_\lambda > \rho_0$ (more initial states fall in). Phase 2 of the algorithm reverses the bending ($\lambda \rightarrow 0$) so the bottom returns to ϕ^* .

Why this matters for the algorithm. Wider mouth = higher probability that a random initial ϕ_0 falls inside the certified basin during the warm-up phase. That is exactly the inequality $p_{\text{entry}}(\text{Meta-MAPG}, T) > p_{\text{entry}}(\text{PG}, T)$.

2 Algorithm and how it connects to F_λ

The proposition is a deterministic statement about F_λ , but it is only useful because the stochastic iteration tracks F_λ in expectation. Here is what the algorithm runs in phase 1.

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input:  $\phi_0 \in \mathbb{R}^d$ , step size  $\alpha_0$ , shaping  $\lambda_c$ , unroll  $L$ , horizon  $N_0$ 
for  $n = 0, 1, \dots, N_0 - 1$ :
    sample batch of trajectories  $\tau_1, \dots, \tau_B$  under joint policy  $\pi_{\phi_n}$ 

    # current term (plain REINFORCE estimator of  $v$ )
     $g_{\text{cur}} \leftarrow \frac{1}{B} \sum_b R_i(\tau_b) S_i(\tau_b)$ 

    # own-learning term (vanishes at Nash but contributes off-Nash)
     $g_{\text{own}} \leftarrow \eta \hat{H}_{ii}(\tau)^\top g_{\text{cur}}$ 

    # peer-learning term (the key Meta-MAPG contribution)
     $g_{\text{peer}} \leftarrow \eta \hat{C}_{ji}(\tau)^\top g_{\text{cur},j}$ 

     $\hat{M} \leftarrow g_{\text{own}} + g_{\text{peer}}$ 
     $\hat{g}_n \leftarrow g_{\text{cur}} + \lambda_c \hat{M}$ 

     $\phi_{n+1} \leftarrow \text{Proj}_{\mathcal{K}}[\phi_n + \alpha_0 \hat{g}_n]$ 

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The bridge to the proposition. The estimator decomposition lemma (`lem:decomp` in the paper) gives

$$\hat{g}_n = F_{\lambda_c}(\phi_n) + \xi_{n+1} + b_n,$$

where ξ_{n+1} is martingale-difference noise and b_n is $O(\beta_n)$ bias. So the iteration tracks the ODE $\dot{\phi} = F_{\lambda_c}(\phi)$. The proposition characterises this ODE near ϕ^* .

What makes the peer term *not* vanish at Nash. The own-learning piece $g_{\text{own}} = \eta \hat{H}_{ii}^\top g_{\text{cur}}$ is a contraction of $g_{\text{cur}} \approx \nabla_{\phi_i} J_i$, which is zero at Nash; so $g_{\text{own}}(\phi^*) = 0$. The peer piece g_{peer} contracts $g_{\text{cur},j} = \nabla_{\phi_j} J_j$, which is also zero at Nash. *However*, the appendix definition of M uses cross-derivatives of value functions: in particular agent j 's gradient enters through agent i 's log-likelihood derivative, so the algebra leaves a term $\eta C_{ji}^\top \nabla_{\phi_j} V_i$ (note the swapped i and j) which is nonzero in general-sum games. That is why $M(\phi^*) \neq 0$ and the fixed point genuinely moves.

3 Notation, fixed once

$v : \mathbb{R}^d \rightarrow \mathbb{R}^d$	policy-gradient field, $v(\phi^*) = 0$
$M : \mathbb{R}^d \rightarrow \mathbb{R}^d$	Meta-MAPG correction, drop L subscript everywhere
$F_\lambda := v + \lambda M$	shaped field at strength $\lambda \geq 0$
ϕ^*	SOS Nash equilibrium of v , drift constant $\mu > 0$, radius r_{att}
ϕ_λ^*	shifted zero of F_λ (built in part (i))
$J_v := Dv(\phi^*)$	Jacobian of v at the unshifted Nash
$J_M := DM(\phi^*)$	Jacobian of M at the unshifted Nash
$J_\lambda := DF_\lambda(\phi_\lambda^*)$	Jacobian of F_λ at the shifted zero
$\text{sym}(A) := (A + A^\top)/2$	symmetric part of a matrix
$\mu := -\lambda_{\max}(\text{sym}(J_v))$	base drift constant (positive by SOS)
$\mu_M := -\lambda_{\max}(\text{sym}(J_M))$	shaping drift constant (assumed positive)
$\mu_\lambda := \mu + \lambda\mu_M$	combined drift constant
L_λ	Lipschitz constant of DF_λ on the working ball
$\rho_\lambda := \mu_\lambda/(2L_\lambda)$	certified SOS radius

Standing regularity. Both v and M are C^2 on a neighbourhood of ϕ^* . This is enough for the IFT (needs C^1 in λ jointly) and for the Taylor remainder (needs C^2).

4 The clean proof

Proposition 1 (Basin expansion). *Fix an SOS Nash equilibrium ϕ^* of v with drift constant $\mu > 0$. Assume $v, M \in C^2$ near ϕ^* . Then for all $\lambda \in [0, \bar{\lambda}]$ with $\bar{\lambda}$ small enough:*

- (i) *there exists a C^1 branch $\lambda \mapsto \phi_\lambda^*$ with $F_\lambda(\phi_\lambda^*) = 0$ and $\|\phi_\lambda^* - \phi^*\| = O(\lambda)$;*
- (ii) *if $\mu_M > 0$, then for every ϕ in the ball $B_{\rho_\lambda}(\phi_\lambda^*)$,*

$$\langle F_\lambda(\phi), \phi - \phi_\lambda^* \rangle \leq -\frac{1}{2}\mu_\lambda \|\phi - \phi_\lambda^*\|^2;$$

- (iii) *the ball $B_{\rho_\lambda}(\phi_\lambda^*)$ strictly contains $B_{r_{\text{att}}}(\phi^*)$ for sufficiently small λ .*

The proof is three lemmas. Each uses one classical tool.

4.1 Tool 1: Implicit Function Theorem

Lemma 1 (Fixed-point shift). *There exist $\bar{\lambda} > 0$ and a C^1 map $\phi^* : [0, \bar{\lambda}] \rightarrow \mathbb{R}^d$, $\lambda \mapsto \phi_\lambda^*$, satisfying $\phi_0^* = \phi^*$ and $F_\lambda(\phi_\lambda^*) = 0$. Moreover*

$$\phi_\lambda^* = \phi^* - \lambda J_v^{-1} M(\phi^*) + O(\lambda^2),$$

so $\|\phi_\lambda^* - \phi^*\| = O(\lambda)$.

Proof. Define $\Phi : \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ by $\Phi(\lambda, \phi) := F_\lambda(\phi) = v(\phi) + \lambda M(\phi)$.

Step 1 (J_v invertible). The SOS condition on ϕ^* states

$$\langle v(\phi), \phi - \phi^* \rangle \leq -\mu \|\phi - \phi^*\|^2 \quad \text{for all } \phi \in B_{r_{\text{att}}}(\phi^*).$$

Substitute $\phi = \phi^* + \varepsilon u$ with $u \in \mathbb{R}^d$ a unit vector. Taylor-expand v :

$$v(\phi^* + \varepsilon u) = v(\phi^*) + \varepsilon J_v u + O(\varepsilon^2) = \varepsilon J_v u + O(\varepsilon^2).$$

The SOS inequality becomes

$$\varepsilon^2 u^\top J_v u + O(\varepsilon^3) \leq -\mu \varepsilon^2.$$

Divide by ε^2 and let $\varepsilon \rightarrow 0$:

$$u^\top J_v u \leq -\mu \quad \text{for all unit } u.$$

So $\text{sym}(J_v) \preceq -\mu I$. Suppose $J_v u = 0$ for some unit u . Then $u^\top J_v u = 0$, contradicting $u^\top J_v u \leq -\mu < 0$. Hence J_v is invertible.

Step 2 (apply IFT). Φ is C^1 jointly because $v, M \in C^2 \supset C^1$. We have $\Phi(0, \phi^*) = v(\phi^*) = 0$ and $\partial_\phi \Phi(0, \phi^*) = J_v$ invertible. The implicit function theorem produces $\bar{\lambda} > 0$ and a unique C^1 branch ϕ_λ^* with $\Phi(\lambda, \phi_\lambda^*) = 0$ on $[0, \bar{\lambda}]$.

Step 3 (first-order expansion). Differentiate $\Phi(\lambda, \phi_\lambda^*) \equiv 0$ in λ :

$$\partial_\lambda \Phi(\lambda, \phi_\lambda^*) + \partial_\phi \Phi(\lambda, \phi_\lambda^*) \cdot \frac{d\phi_\lambda^*}{d\lambda} = 0.$$

At $\lambda = 0$: $\partial_\lambda \Phi = M$ and $\partial_\phi \Phi = J_v$, hence

$$\left. \frac{d\phi_\lambda^*}{d\lambda} \right|_{\lambda=0} = -J_v^{-1} M(\phi^*).$$

Taylor expansion in λ then gives

$$\phi_\lambda^* = \phi^* - \lambda J_v^{-1} M(\phi^*) + O(\lambda^2). \quad \square$$

4.2 Tool 2: Weyl's eigenvalue inequality

Lemma 2 (Linearised drift). *Assume $\mu_M > 0$. There exists $\bar{\lambda}_2 \leq \bar{\lambda}$ such that for all $\lambda \in [0, \bar{\lambda}_2]$ and all $x \in \mathbb{R}^d$,*

$$\langle J_\lambda x, x \rangle \leq -\mu_\lambda \|x\|^2,$$

where $J_\lambda := DF_\lambda(\phi_\lambda^*)$ and $\mu_\lambda = \mu + \lambda \mu_M$.

Proof. **Step 1 (decompose J_λ).** By definition,

$$J_\lambda = Dv(\phi_\lambda^*) + \lambda DM(\phi_\lambda^*).$$

Since $\|\phi_\lambda^* - \phi^*\| = O(\lambda)$ (Lemma 1) and Dv, DM are C^1 ,

$$Dv(\phi_\lambda^*) = J_v + O(\lambda), \quad DM(\phi_\lambda^*) = J_M + O(\lambda).$$

Therefore

$$J_\lambda = J_v + \lambda J_M + O(\lambda^2).$$

The $O(\lambda)$ correction inside $DM(\phi_\lambda^*)$ becomes $O(\lambda^2)$ after multiplication by λ .

Step 2 (extract symmetric part). For any matrix A and any $x \in \mathbb{R}^d$,

$$x^\top A x = \frac{1}{2} x^\top (A + A^\top) x = x^\top \text{sym}(A) x.$$

Applied to J_λ :

$$\langle J_\lambda x, x \rangle = x^\top \text{sym}(J_\lambda) x = x^\top \text{sym}(J_v + \lambda J_M) x + O(\lambda^2) \|x\|^2.$$

Step 3 (Weyl). For real symmetric matrices A, B , Weyl's inequality says

$$\lambda_{\max}(A + B) \leq \lambda_{\max}(A) + \lambda_{\max}(B).$$

Apply to $\text{sym}(J_v) + \lambda \text{sym}(J_M)$:

$$\lambda_{\max}(\text{sym}(J_v + \lambda J_M)) \leq \lambda_{\max}(\text{sym}(J_v)) + \lambda \lambda_{\max}(\text{sym}(J_M)) = -\mu - \lambda \mu_M.$$

Step 4 (Rayleigh). For any symmetric S , $x^\top S x \leq \lambda_{\max}(S) \|x\|^2$. Combine with Step 3:

$$\langle J_\lambda x, x \rangle \leq -(\mu + \lambda \mu_M) \|x\|^2 + O(\lambda^2) \|x\|^2.$$

For λ small, the $O(\lambda^2)$ term is dominated by any positive fraction of μ_λ , so we can absorb it (formally: pick $\bar{\lambda}_2$ such that the residual is ≤ 0 after replacing μ_λ on the right by, say, $0.99\mu_\lambda$; then rename the constant). The clean inequality of the lemma follows. $\square$

4.3 Tool 3: Taylor's theorem with remainder

Lemma 3 (Certified radius). *Assume $\mu_M > 0$ and let $\rho_\lambda = \mu_\lambda/(2L_\lambda)$. For every $\phi \in B_{\rho_\lambda}(\phi_\lambda^*)$,*

$$\langle F_\lambda(\phi), \phi - \phi_\lambda^* \rangle \leq -\frac{1}{2}\mu_\lambda \|\phi - \phi_\lambda^*\|^2.$$

Proof. Write $x := \phi - \phi_\lambda^*$. Since $F_\lambda \in C^2$ on a neighbourhood of ϕ_λ^* , Taylor's theorem with integral remainder gives

$$F_\lambda(\phi) = F_\lambda(\phi_\lambda^*) + J_\lambda x + R_\lambda(x), \quad \|R_\lambda(x)\| \leq \frac{1}{2}L_\lambda \|x\|^2.$$

By Lemma 1, $F_\lambda(\phi_\lambda^*) = 0$. Take inner product with x :

$$\langle F_\lambda(\phi), x \rangle = \langle J_\lambda x, x \rangle + \langle R_\lambda(x), x \rangle.$$

Bound the first term by Lemma 2 and the second by Cauchy-Schwarz:

$$\langle F_\lambda(\phi), x \rangle \leq -\mu_\lambda \|x\|^2 + \|R_\lambda(x)\| \cdot \|x\| \leq -\mu_\lambda \|x\|^2 + \frac{1}{2}L_\lambda \|x\|^3.$$

For $\|x\| \leq \rho_\lambda = \mu_\lambda/(2L_\lambda)$,

$$\frac{1}{2}L_\lambda \|x\|^3 = \frac{1}{2}L_\lambda \|x\| \cdot \|x\|^2 \leq \frac{1}{2}L_\lambda \cdot \frac{\mu_\lambda}{2L_\lambda} \|x\|^2 = \frac{\mu_\lambda}{4} \|x\|^2.$$

Therefore

$$\langle F_\lambda(\phi), x \rangle \leq -\mu_\lambda \|x\|^2 + \frac{\mu_\lambda}{4} \|x\|^2 = -\frac{3\mu_\lambda}{4} \|x\|^2 \leq -\frac{1}{2}\mu_\lambda \|x\|^2. \quad \square$$

4.4 Putting it together

Proof of the proposition. Part (i) is Lemma 1. Part (ii) is Lemma 3. For part (iii), compare the certified radius $\rho_\lambda = (\mu + \lambda \mu_M)/(2L_\lambda)$ to r_{att} . As $\lambda \rightarrow 0^+$, $L_\lambda \rightarrow L_0$ by smoothness of M , so $\rho_\lambda \rightarrow \mu/(2L_0)$. The latter is at least r_{att} whenever the original SOS ball is the maximal one; if it is strictly smaller, $\rho_\lambda > r_{\text{att}}$ already at $\lambda = 0$. In either case, for $\lambda > 0$ sufficiently small, $\rho_\lambda > r_{\text{att}} + c_{\text{shift}}\lambda$ where c_{shift} is the constant of Lemma 1, which means the new ball $B_{\rho_\lambda}(\phi_\lambda^*)$ strictly contains the old ball $B_{r_{\text{att}}}(\phi^*)$. $\square$

5 Why this is elegant

Three classical tools, one per part, applied in the natural order:

- **IFT** answers “does the zero move and where?” Solves $F_\lambda(\phi) = 0$ for ϕ as a function of λ .
- **Weyl** answers “how does the slope change?” Adds eigenvalue bounds of two symmetric matrices.
- **Taylor** answers “how far does the linear bound extend?” Trades cubic remainder against linear contraction.

Nothing clever is needed. Each step is the obvious move once the previous one is in place. The art is in choosing the working point $(\phi_\lambda^*, \text{not } \phi^*)$ so that the linearisation has zero constant term.

6 Lean4 guidance

A full Mathlib formalisation is heavy because Mathlib has the IFT and Taylor’s theorem and an eigenvalue API but stitching them around a custom F_λ takes work. The skeleton in `BasinExpansion.lean` is a checklist: each `sorry` is one human-proof step. Use it like this.

- **Sorry 1** (`J_v_invertible`): proves J_v is non-singular under SOS. The proof is the contra-positive argument from Lemma 1 Step 1.
- **Sorry 2** (`fixed_point_branch`): invokes `ImplicitFunctionTheorem.implicitFunctionData` from `Mathlib.Analysis.Calculus.ImplicitFunction`. Inputs: $\Phi(\lambda, \phi) = v(\phi) + \lambda M(\phi)$, base point $(0, \phi^*)$, invertibility from **Sorry 1**.
- **Sorry 3** (`shift_first_order`): differentiate $\Phi(\lambda, \phi_\lambda^*) \equiv 0$ and invert J_v .
- **Sorry 4** (`jacobian_decomposition`): uses C^1 continuity of Dv, DM to write $J_\lambda = J_v + \lambda J_M + O(\lambda^2)$.
- **Sorry 5** (`weyl_bound`): standard Weyl on Hermitian matrices. Mathlib’s `Matrix.IsHermitian.eigenvalue` and the inequality `Matrix.eigenvalues_add_le` (search for `Weyl`) cover this.
- **Sorry 6** (`rayleigh_bound`): Mathlib has `Matrix.IsHermitian.inner_le_eigenvalues_mul_norm_sq` or analogous.
- **Sorry 7** (`taylor_remainder`): use `HasFTaylorSeriesUpToOn / taylorWithinEval` or the simpler `HasFDerivAt.le` with C^2 bounds.
- **Sorry 8** (`cubic_dominates`): pure arithmetic on $\|x\| \leq \rho_\lambda$.

The point of the Lean skeleton: it forces you to name each object precisely (no notation drift), exhibit each input, and check that nothing is silently assumed. Even without filling in all `sorry`s, walking the structure from start to end catches gaps. Three places worth checking against the human proof:

1. In **Sorry 4**, the step “ $O(\lambda)$ inside DM becomes $O(\lambda^2)$ after multiplication by λ ” is honest only because we evaluate DM at a moving base point. If you write it at ϕ^* instead, you lose this gain.

2. In **Sorry 7**, L_λ depends on the ball where you take the Lipschitz constant of DF_λ . The ball is $B_{\rho_\lambda}(\phi_\lambda^*)$, which depends on ρ_λ , which depends on L_λ . This is not circular: pick any fixed ball B around ϕ^* that contains $B_{\rho_\lambda}(\phi_\lambda^*)$ for all small λ , take L as the Lipschitz constant on B , and use it for every λ . The Lean skeleton makes this choice explicit; the paper hides it.
3. In **Sorry 8**, the constant 2 in $\rho_\lambda = \mu_\lambda/(2L_\lambda)$ is exactly the “half budget” choice. With factor k instead of 2, the certified drift becomes $-(1 - 1/(2k))\mu_\lambda\|x\|^2$. Choosing $k = 2$ gives the cleanest constant $-\frac{3}{4}\mu_\lambda$ and the comfortable $-\frac{1}{2}\mu_\lambda$ stated in the proposition.